rezki ::= engineer | educator

# Rezki Boureghdad

Back-end & Data Architect · Computer Science Educator

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## PROFILE

Software engineer focused on back-end and data architecture, and a computer-science educator. At Solrus, an Algiers software studio, I design the data models, APIs and integrations behind client production platforms. I also teach at Université M'Hamed Bougara, where I have covered 12 subjects, from algorithms and databases to compilation and semantics, at Bachelor's and Master's level. I take the same approach in both roles: get the structure right, make it measurable, and explain it clearly.

## EXPERIENCE

### Data Specialist, Structure & Architecture

2025 – Present

Solrus — software studio for AI, cloud and secure systems · Algiers, remote-first

- Own back-end and data architecture for client platforms: domain modelling, relational schemas and migrations, API contracts, and third-party integrations.
- Build event-driven and async Python services (FastAPI, SQLAlchemy 2, Kafka, Redis, Celery) and tune them for latency and reliability.
- Security in the design from the start: role-based access, 2FA, audit trails, data-residency compliance, hardening before audits.
- Test-first delivery with pytest against real PostgreSQL and end-to-end Playwright suites. Work closely with front-end, AI and product teammates.

> Most client projects are under NDA. Names and details are not disclosed.

### Freelance Full-Stack Developer (auto-entrepreneur)

May 2021 – Present

Independent · web, mobile and AI products

- Run projects end to end for local businesses and community organisations: scoping, proposal, clickable prototype, build and deployment.
- Member of the team behind **Zakat.business**, an award-winning non-profit startup project.

### Teaching Assistant — Computer Science

Nov 2024 – Present

Université M'Hamed Bougara, Boumerdès

- 12 subjects taught so far to Software Engineering and LMD students, Bachelor's and Master's level.
- Prepare and run tutorials and labs, write exercises and assessments, and help students connect theory to hands-on practice.
- Adjust my teaching for each audience, from first programming courses to formal semantics for Master's students.

### Development Mentor

Oct 2024 – Sep 2026

DevUp Club, Université M'Hamed Bougara

- Designed and ran practical software-engineering workshops for students, with hands-on material that participants applied to real projects.

## EDUCATION

**MSc Software Engineering & Data Science**
UMBB · Jan 2024

**BSc Computer Science**
UMBB · Jan 2022

## SUBJECTS TAUGHT

Algorithms & Data Structures

Databases

Programming Languages (intro → advanced)

Compilation

Semantics

+ 7 other CS modules

## CERTIFICATIONS

**NVIDIA**
Fundamentals of Deep Learning · Intro to Machine Learning

**Neo4j**
Graph Data Science · Fundamentals · Graph Data Modeling

**EF SET English**
73/100 — C2 Proficient

## LANGUAGES

**Arabic**
Native

**English**
C2 (EF SET)

**French**
Professional working

## SELECTED PROJECTS

Public, open and personal work only.

### Forsa / Aamal ATS — recruitment platform

Solrus public case study

- Connects companies, independent recruiters and job seekers in Algeria. Semantic job matching with sentence-transformer embeddings and location search with PostGIS.
- WebSocket chat, community forum, FCM push notifications, SlickPay subscriptions, JWT/Argon2 auth. Flutter mobile app and admin panel.

FastAPI · PostgreSQL/PostGIS · Flutter · Docker

### e-motion — back-office for cash-on-delivery shops

Own product

- Orders, stock, couriers and expenses, with real profit per order. Weighted-average purchase cost, stock that can never go below zero, alerts for customers who often cancel or return, and an audit log.
- Excel import, packing slips, English/French interface. Runs on free hosting tiers.

Next.js · TypeScript · Supabase · Vercel

### DocMCP — local PDF search server for AI tools

Open source

- Ingests PDF folders into ChromaDB and serves them to any MCP client (Kiro, Claude Code) as semantic-search tools. Fully local: ONNX / sentence-transformer embeddings, LLM on Ollama.
- Recursive Language Model reasoning: a local LLM explores the full documents in a Python REPL instead of relying on top-k chunks. Web UI and one-command Docker setup.

Python · Node.js · MCP · ChromaDB · Ollama · Docker · github.com/V0idSeeker/RAG-MCP

### Ahl Quran — multi-tenant school manager

Community

- Study circles, attendance, student progress, messaging and reports, separated by organisation. Web and mobile apps.

FastAPI · PostgreSQL · Flutter/GetX · Docker

### 7alkaty — mosque course & imam-following app

Mobile

- REST API documented with Swagger, role-based access, scheduled push notifications, media storage, and a React Native client.

Node/Express · MongoDB · Expo

### Forest-fire detection & reporting

Master's thesis

- Co-designed a three-part system: a citizen reporting app, a data platform and alerting. Detection models built with YOLO and Faster R-CNN.

Python · YOLO · R-CNN · Flutter · Firebase

### claim-bot — migration-number Discord bot

Own tool

- Reserves the next migration number before the file exists, so developers never collide. Merges committed files from the GitHub API with open claims, and checks the Alembic chain for extra heads, broken links and cycles.

Python · discord.py · GitHub GraphQL · Alembic

### ISO 27001 skill for Claude Code

Tooling

- Checks a codebase against ISO/IEC 27001:2022, plans the rollout step by step, and checks every edit with a scanner hook.

Python · Security compliance · AI tooling

## SKILLS

**Back-end**
Python, FastAPI, Django/DRF, SQLAlchemy 2 (async), Alembic, Pydantic, Celery, Node/Express

**Data**
PostgreSQL/PostGIS, Redis, Kafka, MongoDB, Neo4j, Elasticsearch, Supabase

**AI / ML**
RAG, MCP servers, LangGraph, FAISS/Chroma, embeddings & reranking, LLM APIs, YOLO/R-CNN

**Front & mobile**
TypeScript, Next.js, React, Tailwind, Flutter/Dart, React Native

**Ops & quality**
Docker, nginx, GitHub Actions, Vercel, Grafana/Loki, pytest, Playwright

**Security**
JWT, Argon2, OAuth2, TOTP 2FA, ISO 27001, data residency

**Also**
Java, C++, SQL; contract-first API design, decision records, AI-assisted development

## CS FOUNDATIONS

Algorithms & complexity

Compilers & formal languages

Formal methods

Deep learning

Programming-language semantics

Database theory & graph databases